

 HARD

 ADVANCED MATH

Advanced Math

12

10 questions · ~15 min

Q1

GRID-IN ADVANCED MATH

$$\sqrt{k-x} = 58-x$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The function g is defined by $gx = \frac{x}{a} - 14$, where $a < 0$. What is the product of $g15a$ and $g7a$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$2x^2 - 4x = t$$

In the equation above, t is a constant. If the equation has no real solutions, which of the following could be the value of t ?

- A. -3
- B. -1
- C. 1
- D. 3

Function f is defined by $f(x) = x^3 + 6x^2 + 5x + 1$. Function g is defined by $g(x) = f(x) - 1$. The graph of $y = g(x)$ in the xy -plane has x -intercepts at $(a, 0)$, $(b, 0)$, and $(c, 0)$, where a , b , and c are distinct constants. What is the value of $a + b + c$?

- A. -15
- B. -9
- C. 11
- D. 15

$$57x^2 + (57b + a)x + ab = 0$$

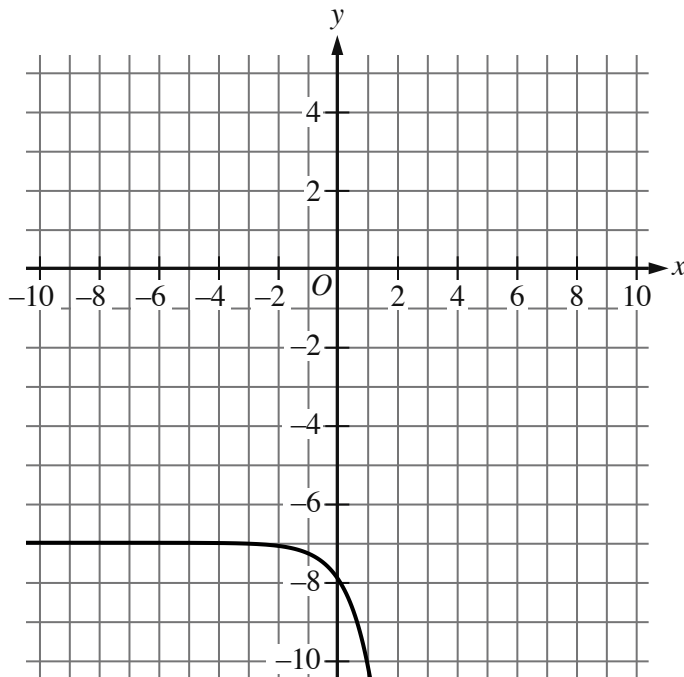
In the given equation, a and b are positive constants. The product of the solutions to the given equation is kab , where k is a constant. What is the value of k ?

A. $\frac{1}{57}$

B. $\frac{1}{19}$

C. 1

D. 57



- Moving from left to right, the curve passes from quadrant 3 to quadrant 4.
- As x decreases, the curve approaches line y equals negative 7.
- The curve passes through the following approximate points:
 - (negative 1 comma negative 7.3)
 - (0 comma negative 7.9)

The graph of $y = fx$ is shown, where $fx = ab^x + c$, and a , b , and c are constants. For how many values of x does $fx = 0$?

- A. Three
- B. Two
- C. One
- D. Zero

$$64x^2 - 16a + 4bx + ab = 0$$

In the given equation, a and b are positive constants. The sum of the solutions to the given equation is $k4a + b$, where k is a constant. What is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$gx = \frac{x^2 - x - a}{x^3 - x - b}$$

The function g is defined by the given equation, where a and b are constants. In the xy -plane, the graph of $y = gx$ passes through the point $(0, 2)$, and $g(-2) = 0$. What is the value of b ?

- A. 23
- B. 22
- C. -22
- D. -23

$$D = T - \frac{9}{25}(100 - H)$$

The formula above can be used to approximate the dew point D , in degrees Fahrenheit, given the temperature T , in degrees Fahrenheit, and the relative humidity of H percent, where $H > 50$. Which of the following expresses the relative humidity in terms of the temperature and the dew point?

A. $H = \frac{25}{9}(D - T) + 100$

B. $H = \frac{25}{9}(D - T) - 100$

C. $H = \frac{25}{9}(D + T) + 100$

D. $H = \frac{25}{9}(D + T) - 100$

$$h(x) = -16x^2 + 100x + 10$$

The quadratic function above models the height above the ground h , in feet, of a projectile x seconds after it had been launched vertically. If $y = h(x)$ is graphed in the xy -plane, which of the following represents the real-life meaning of the positive x -intercept of the graph?

- A. The initial height of the projectile
- B. The maximum height of the projectile
- C. The time at which the projectile reaches its maximum height
- D. The time at which the projectile hits the ground